

CLASS XII · LECTURE 7

C Programs for the Examination

Option I · A practical lecture · Type every program, then write it from memory

By the end of this lecture you can

- Write, from a blank page, the table, factorial, greater-of-two and triangle programs.
- Trace the three integer expressions set with $a = 2$, $b = 5$, $c = 10$.
- Convert the standard algebraic formulas into C.

How to practise a program question

Close the notes. Write the program. Then type it, or trace it by hand if you are not at a machine. A program that you have only read will lose its braces under exam pressure. The four programs below are the set the paper keeps returning to. Database questions are in Lectures 12 to 14 and will sit in the same Section B.

1. Multiplication table

Read n . Loop i from 1 to 10. Print n , i and n times i on each line. That is Program 4.1. If the question says “inputted number”, the `scanf` is compulsory. A table of a fixed 5, with no input, does not answer it.

2. Factorial

Program 4.2. Under the program, write the trace for 4: fact becomes 1, 2, 6, 24. The hint in the question is there so that a correct trace confirms your loop. Starting fact at 0 ruins the answer, because every product stays 0.

3. Greater of two numbers, using a function

Program 5.1. The question says “using function”. A program that compares inside main and never defines a second function misses the point. The function must return the value. Printing inside the function can be accepted, but returning is the cleaner match to “a function that returns”.

4. The two patterns

Program 4.4 prints the triangle of numbers. Program 4.5 prints the number and its square. Write the output beside the program for one mark of confidence: the triangle’s third line is 123, and the square of 4 is 16.

5. Expressions and formulas

From Lecture 2, with $a = 2$, $b = 5$, $c = 10$: the three results are 6000, 22 and 101. Show the working.

Formula conversions: use 0.5 for a half, stars between every factor, `fabs` for an absolute value, and $r * r$ for a square. Include `math.h` when you call `fabs`.

EXAM TIP What a marker looks for

The header. The types. The `&` in `scanf`. The loop limits. The place where the answer is printed. A missing brace is serious. A missing comment is not. Do not spend ten minutes decorating the program.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Short answers

S1. Write a program that reads a number and prints its multiplication table.

S2. Write a nested loop that prints 1, then 12, then 123, then 1234, then 12345.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

S1. See Program 4.1. Include `stdio.h`, read `n` with `scanf("%d", &n)`, and for `i` from 1 to 10 print `n`, `i` and `n * i`. End main with `return 0`.

S2. See Program 4.4. Outer loop row from 1 to 5. Inner loop col from 1 to row, printing col with no newline. After the inner loop, print a newline.